Banking AI Agent Team Roadmap

A proposed 12-day delivery plan for five teammates | Prepared 18 October 2026

Priority: connect the working modules early, deliver three reliable banking scenarios, then finish one bounded connected workflow. Do not wait for each person to finish all standalone features before integrating.

Day 1 means the day the team adopts this plan. The submission date is not confirmed here; compress the schedule if needed and protect the final two days for submission and rehearsal. All operations use synthetic data and simulated banking execution.

Current code baseline

Owner	Present in the reviewed checkout	Main gap
P1 Agent	Gemini/mock providers, typed proposals, clarification and tests	Multi-turn state handoff and backend integration
P2 App - Marvel	FastAPI, database, transfer approval/MFA and tests	Real adapters, subscriptions, frontend and deployment
P3 Finance	Totals, categories, anomalies, diagnostics and tests	Data handoff and integrated financial verification
P4 Security	Only a placeholder is merged	Review their branch and integrate the real policy
P5 Subscriptions	Cancellation eligibility and two tests	Detector, evidence/confidence and ID mapping

Combined suite: 148 tests passed with one existing SDK warning. This is not evidence that the entire app works end to end. Unmerged work, particularly P4's, is not assessed here.

1 Interface agreement and blockers | Days 1-2

Involved: All five; P2 coordinates.

P1 + P2: Agree catalog/context, clarification state and correlated tool-result examples.

P2 + maintainers: Use canonical recipient_id and selected account IDs; remove fallback recipients. Resolve API/approval metadata differences through approved versioned contracts.

P2 + P3/P4/P5: Agree China-time/history filtering, risk boundaries, candidate-to-subscription mapping, and one shared synthetic demo dataset.

Complete when: Every handoff has a concrete input/output example and an owner for unresolved changes.

2 First integrated flow and basic UI | Days 3-4

Involved: P1 + P2; others continue their modules.

P1: Build a small conversation harness and basic multi-turn clarification while agreeing the backend handoff.

P2: Make the adapter switch functional, call the real agent, execute read-only balance lookups, and start chat/balance/loading/error UI.

Complete when: An app user receives a database-grounded balance reply and can answer a clarification in the correct conversation.

Team delivery milestones

Integration targets and acceptance checks

3 Three standalone scenarios | Days 5-6

Involved: All five.

Spending - P1/P2/P3: P2 retrieves selected-account transactions plus required history; P3 calculates; P1 explains totals, categories and anomalies.

Transfers - P1/P2/P4: P1 proposes; P4 evaluates; P2 confirms, requests simulated MFA where required, and executes safely.

Subscriptions - P1/P2/P4/P5: P5 detects recurring charges and assesses eligibility; P1 explains evidence; P2 persists subscription state and routes cancellation separately from transfers.

Complete when: All three work through the app, each with at least one clarification or rejection case. Read-only requests complete without transfer confirmation.

4 Connected workflow and recovery | Days 7-8

Involved: P1 + P2 + P4 + P5; P3 checks financial wording.

Sequence: Find recurring charge; clarify selection; propose cancellation; obtain approval; receive trusted cancellation completion; then propose a separate, agreed transfer with its own approval/MFA.

Responsibilities: P1 proposes the next step; P2 stores workflow state and executes; P4 checks each mutation; P5 supplies evidence and eligibility; P3 checks savings calculations.

Recovery: Support never mind, changed amount, rejected cancellation and failed execution. Projected future savings are not a new account credit.

Complete when: Dependencies are respected and the sequence can stop safely before a later action.

5 Security deployment and freeze | Days 9-10

Involved: P4 leads security; P2 leads deployment; everyone fixes owned code.

Security: Test ownership violations, prompt injection, stale results, tampered approvals, expiry, repeated/concurrent confirmations, and current lock/funds/policy checks at execution.

Persistence: Verify daily-transfer reset boundaries, genuine audit records, stable replay results, and transaction safety on the chosen deployment database.

Operations: Complete start/stop, synthetic seed/reset and deployment commands. Test provider failure and explicitly labelled mock mode; freeze new features.

Complete when: Another teammate can start the project and run all agreed scenarios from the documentation.

6 Submission and rehearsal | Days 11-12

Involved: All five.

Deliverables: Project brief up to 200 words; technical architecture/algorithm/security documentation; source, README, deployment guide and tests; security self-assessment and known risks.

Presentation: Defence presentation within 10 minutes and demo video within 5 minutes. Rehearse from a clean synthetic state: normal flow, confirmation/MFA, and one safely handled failure.

Complete when: Submission materials are complete and the team can reproduce and explain the final demonstration.

Individual milestones

P1 Agent P2 App P3 Finance

P1 Agent

Days 1-4 | With P2: Finish multiple clarification turns, corrections/interruption, conversation isolation, and result-to-request correlation. Keep process_message stateless; the harness/backend owns state. Agree context and result examples with Marvel.

Days 5-8 | With P2/P3/P4/P5: Explain real finance reports and subscription findings. Add the bounded dependent workflow and handle pending, failed, rejected and completed outcomes without inventing success.

Days 9-12 | With P4 and the team: Fix agent-side adversarial failures. Provide prompting/validation architecture, limitations, reproducible examples and demo narration.

Deliverable: A tested conversational agent integrated into the app; all source changes remain in agent/ and docs/agent/. Do not duplicate backend SQL, persistence or execution.

P2 App and backend - Marvel

Days 1-4 | With all owners: Align shared schemas, IDs and approval metadata; preserve canonical arguments; implement real/mock adapter selection, tool dispatch and conversation persistence. Start the frontend now rather than after all backend work.

Days 5-8 | With P1/P3/P4/P5: Connect real modules. Add verified recipient/subscription records and consistent synthetic seed data. Finish balance/report/subscription UI, confirmation and MFA, and dedicated cancellation execution.

Days 9-12 | With P4: Protect concurrent confirmation/idempotency, execution-time checks, expiry and audit. Finish the deployment database setup, migrations, seed/reset, Docker/startup instructions and end-to-end tests.

Deliverable: A reproducible working application. P2 is the composition root and database owner; policy permission alone is never execution authorization.

P3 Finance analytics

Days 1-4 | With P2 and P1: Confirm account scope, three-month anomaly history and Asia/Shanghai month boundaries. Supply expected report examples for integration. Keep agreed outbound-transfer/refund rules explicit.

Days 5-8 | With P1/P2: Verify real app totals, category labels, anomaly explanations, empty periods and timestamp boundaries. Confirm how optional diagnostic notes reach the UI; no notes field is added without team agreement.

Days 9-12 | With the team: Own financial acceptance checks and expected numeric demo results. Check projected-savings wording and prepare finance evidence for the documentation/presentation.

Deliverable: Deterministic reports that match the displayed account/period, plus tested expected figures. P3 does not own database queries or money execution.

Individual milestones and team rules

P4 Security P5 Subscriptions Working agreements

P4 Security and risk

Days 1-4 | With P2/P1: Merge or review the real policy implementation. Confirm cumulative daily thresholds, exactly-CNY-1000 behavior, read-only versus mutation locks, invalid-input handling, and reason codes/messages.

Days 5-8 | With P2/P1/P5: Check enforcement at execution and each connected-workflow checkpoint. Distinguish action failure/lock from account-level safety lock, and define repeated-failure behavior.

Days 9-12 | With everyone: Lead adversarial tests: ownership, injection, altered payloads, approval reuse, expiry, concurrent execution and sensitive error/log handling. Write the security self-assessment and known-risk list.

Deliverable: A pure fail-closed policy module and security evidence; P2 enforces it and owns approval/MFA persistence/execution.

P5 Subscriptions

Days 1-4 | With P2: Implement recurring-charge detection with confidence and transaction evidence. Define stable candidate IDs and mapping to stored subscriptions; confirm supported-cancellation metadata and active-state checks.

Days 5-8 | With P1/P2/P4: Integrate detection, listing and cancellation eligibility. Test weak evidence, irregular charges, unsupported merchants and already-cancelled subscriptions. Separate candidates from verified active subscriptions.

Days 9-12 | With the team: Verify evidence and stored status after simulated execution/reset. Document confidence/eligibility limits and supply reliable subscription demo examples.

Deliverable: A tested detector and eligibility function with concrete input/output examples. P2 performs simulated cancellation; P5 never executes it.

Working agreements for the remaining days

Branches and ownership: Continue on feature branches, edit only owned areas, and merge small working PRs regularly. Shared changes require maintainer agreement and versioning.

Daily integration check: Briefly show what runs through the app, identify the current blocker, name its owner and agree the next check. Tests passing individually are not an integrated demo.

Avoid the P2 bottleneck: P3/P4/P5 own domain fixtures, acceptance checks and submission sections. P1 owns agent examples and AI-design material; P2 should not write every test and document.

Scope control: Deliver three strong scenarios plus one bounded connected workflow. Defer real banks, additional banking products and cosmetic features. Stubs remain clearly labelled until replaced.

Immediate next action: P1 starts milestone three in coordination with P2 adapter work. P2 starts basic UI; P3 confirms report inputs; P4 exposes the real policy; P5 delivers the detector.